Back

Time to Practice: DBSCAN Clustering
Graded Assignment • 15 min

Due Oct 21, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. A data scientist is using DBSCAN for clustering a dataset with complex groups of points. They notice that all points are being labeled as outliers (noise). What is the most likely reason for this observation? 1 / 1 point

- The dataset is too large
- The value of the epsilon (eps) parameter is too low
- The value of the 'min samples' parameter is too high.

Nice work

If the eps value is set too small, the neighborhood radius around each point is very limited, causing no points to be within the neighborhood of others. This results in all points being labeled as noise.

2. What is a major advantage of using DBSCAN over k-means clustering? 1 / 1 point

- DBSCAN does not require you to specify the number of clusters in advance
- DBSCAN is more accurate with non-numeric data
- DBSCAN performs a much faster update of cluster centroids

Nice work

DBSCAN does not require the number of clusters to be specified in advance, unlike k-means, which requires this information. This makes DBSCAN more flexible and applicable to datasets where the number of clusters is not known beforehand.

3. In the DBSCAN algorithm, what distinguishes a boundary point from a core point? 1 / 1 point

- Lack of sufficient close neighbors
- Closer to cluster centroids
- Greater distance from noise points
- Higher density than core points

Nice work

Boundary points are within the distance eps from core points but do not have enough close neighbors to be considered core points themselves.

4. Why is DBSCAN effective at finding complex cluster shapes, compared to methods like k-means clustering? 1 / 1 point

- It requires fewer parameters
- It identifies local dense regions while making minimal assumptions about the global cluster shape.
- It finds irregular regions efficiently by removing noise points

Nice work

5. What impact does increasing the 'min samples' parameter have on the clustering outcomes from DBSCAN? 1 / 1 point

- DBSCAN will find clusters that are more dense
- DBSCAN will find clusters that are more sparse
- There will typically be no change in the nature of the clusters that are found

Nice work

Increasing the 'min samples' parameter raises the threshold for the number of nearby points needed for a point to be considered a core sample. This results in a stricter definition of density and more dense clusters.